

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (IT) Examination

May 2013

IT404 Language Processor

Date: 11.05.2013, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I

- Q - 1 (a) What is language processor? Explain types of language processor. [04]
- (b) Differentiate between [03]
- 1) Compiler and Interpreter
 - 2) Token and Lexeme
- Q - 2 (a) List out the different phases of compiler. Explain significance of each phase of compiler with an example. [08]
- (b) Draw DFA for following RE [04]
- $(111+ 100)^*0$
- (c) Explain synthesized attribute and inherited attribute. [02]

OR

- Q - 2 (a) Draw NFA- $\wedge$ for following regular expression [08]
- $(a/b)^*abb$
- Construct DFA from NFA. Also prepare transition table.
- (b) Explain two pass assembly in detail. [04]
- (c) List out the different types of errors in compiler and give an example of each. [02]
- Q - 3 (a) Explain the role of the lexical analyzer. [06]
- (b) What is symbol table? Why it is used? Which phase of compiler set the attributes of token in a symbol table? [05]
- (c) Explain the single pass and multi pass compiler. [03]

OR

- Q - 3 (a) Explain Error Recovery Strategies in detail with suitable example. [06]
- (b) What is activation record? Explain clearly the components of an activation record. [05]
- (c) Check whether the below grammar is Ambiguous or not? Which language is generated from this grammar? [03]

$$S \rightarrow aSbS \mid bSa \mid \epsilon$$

SECTION – II

Q - 4 (a) Define following terms: [08]

- 1) Left recursion
- 2) Handle pruning
- 3) Three address code
- 4) S-R conflicts

Q - 5 (a) Generate Canonical LR(0) item set for following grammar [05]

$S \rightarrow L = R$

$S \rightarrow R$

$L \rightarrow *R$

$L \rightarrow id$

$R \rightarrow L$

(b) Compute FIRST and FOLLOW set for following [04]

$E \rightarrow TE'$

$E' \rightarrow +TE' \mid \epsilon$

$T \rightarrow FT'$

$T' \rightarrow *FT' \mid \epsilon$

$F \rightarrow (E) \mid id$

(c) Write Quadruples for [04]
 $(a+b)*(c+d)-(a+b+c)$

OR

Q - 5 (a) Check following grammar is LL(1) or not. [05]

$S' \rightarrow S\#$

$S \rightarrow aSA \mid \epsilon$

$A \rightarrow bS \mid \epsilon$

(b) What is operator grammar? Write advantages and disadvantages of operator precedence parsing. [04]

(c) Perform the Left factoring of following Grammar [04]

$A \rightarrow ad \mid a \mid ab \mid abc \mid b$

Q - 6 (a) List out the different code optimization techniques. [02]

(b) Attempt any Three. [12]

- 1) Explain the functions of linker and loader.
- 2) Compare static memory allocation with dynamic memory allocation.
- 3) Preprocessor
- 4) Compare Top Down and Bottom Up parser.
